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-- HR Analytics SQL Project  
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```

```
-- 1. Database Schema Setup
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```

```
CREATE TABLE hrdata (  
    emp_no INT8 PRIMARY KEY,  
    gender VARCHAR(50) NOT NULL,  
    marital_status VARCHAR(50),  
    age_band VARCHAR(50),  
    age INT8,  
    department VARCHAR(50),  
    education VARCHAR(50),  
    education_field VARCHAR(50),  
    job_role VARCHAR(50),  
    business_travel VARCHAR(50),  
    employee_count INT8,  
    attrition VARCHAR(50),  
    attrition_label VARCHAR(50),  
    job_satisfaction INT8,  
    active_employee INT8  
);
```

```
SELECT *  
FROM hrdata;
```

```
-- 2. Key Performance Indicators (KPIs)
```

```
-- =====
```

```
-- Employee Count:
```

```
SELECT sum(employee_count) AS Employee_Count  
FROM hrdata;
```

```
-- Attrition Count:
```

```
SELECT count(attrition)  
FROM hrdata  
WHERE attrition = 'Yes';
```

```
-- Attrition Rate:
```

```
SELECT round(  
    ((SELECT count(attrition) FROM hrdata WHERE attrition = 'Yes') / sum(employee_count) *  
    100), 2  
)  
FROM hrdata;
```

-- Average Age:

```
SELECT round(avg(age), 0)
FROM hrdata;
```

-- Active Male Employees:

```
SELECT sum(employee_count) - (
    SELECT count(attrition)
    FROM hrdata
    WHERE attrition = 'Yes' AND gender = 'Male'
)
FROM hrdata
WHERE gender = 'Male';
```

-- 3. Employee Distribution

-- =====

-- Count based on department (Sales)

```
SELECT sum(employee_count)
FROM hrdata
WHERE department = 'Sales';
```

-- Count based on department (R&D)

```
SELECT sum(employee_count)
FROM hrdata
WHERE department = 'R&D';
```

-- Count based on education (High School)

```
SELECT sum(employee_count)
FROM hrdata
WHERE education = 'High School';
```

-- Count based on education field (Medical)

```
SELECT sum(employee_count)
FROM hrdata
WHERE education_field = 'Medical';
```

-- 4. Attrition Breakdown

-- =====

-- Attrition for Doctoral Degree

```
SELECT count(attrition)
FROM hrdata
WHERE attrition = 'Yes' AND education = 'Doctoral Degree';
```

-- Attrition based on multiple criteria

```

SELECT count(attrition)
FROM hrdata
WHERE attrition = 'Yes'
      AND education_field = 'Medical'
      AND department = 'R&D'
      AND education = 'High School';

```

-- Department-wise Attrition (Volume and Percentage)

```

SELECT department,
       count(attrition),
       round((CAST(count(attrition) AS numeric) /
              (SELECT count(attrition) FROM hrdata WHERE attrition = 'Yes')) * 100, 2) AS pct
FROM hrdata
WHERE attrition = 'Yes'
GROUP BY department
ORDER BY count(attrition) DESC;

```

-- Gender-wise Attrition (Volume and Percentage)

```

SELECT gender,
       count(attrition),
       (count(attrition) / (SELECT count(attrition) FROM hrdata WHERE attrition = 'Yes')) * 100
FROM hrdata
WHERE attrition = 'Yes'
GROUP BY gender
ORDER BY count(attrition) DESC;

```

-- 5. Advanced Analysis: Job Satisfaction Crosstab

-- =====

-- Job Satisfaction Rating

CREATE EXTENSION IF NOT EXISTS tablefunc;

```

SELECT *
FROM crosstab(
  'SELECT job_role, job_satisfaction, sum(employee_count)
   FROM hrdata
   GROUP BY job_role, job_satisfaction
   ORDER BY job_role, job_satisfaction'
) AS ct(job_role varchar(50), one numeric, two numeric, three numeric, four numeric)
ORDER BY job_role;

```